

Lecture 19 Scribe

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Below is a **Lecture 19 Scribe** based strictly on the visible slide structure and standard definitions consistent with the listed topics. Where slide text is not readable, only **canonical definitions aligned with the slide headings** are used, without adding extra concepts beyond scope.

Lecture 19 Scribe: N-Variate Joint PDF / PMF / CDF

Course: CSE400 – Fundamentals of Probability in Computing

Instructor: Dhaval Patel, PhD

Lecture: 19

Topic: Joint Distribution of N Random Variables

1. Joint Distribution of N Random Variables – Definition

Let $X_1, X_2, \dots, X_n$ be n random variables defined on the same probability space.

Definition (Joint CDF)

The **joint cumulative distribution function (CDF)** is defined as:

$$F_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n) = P(X_1 \leq x_1, X_2 \leq x_2, \dots, X_n \leq x_n)$$

Step-by-step reasoning

1. Consider all random variables simultaneously.
2. The event is the intersection:

$$\{X_1 \leq x_1\} \cap \{X_2 \leq x_2\} \cap \dots \cap \{X_n \leq x_n\}$$

3. The joint CDF is the probability of this joint event.

2. Joint PMF (Discrete Case)

Definition

For discrete random variables:

$$p_{X_1, \dots, X_n}(x_1, \dots, x_n) = P(X_1 = x_1, \dots, X_n = x_n)$$

Properties

1. Non-negativity:

$$p(\cdot) \geq 0$$

2. Normalization:

$$\sum_{x_1} \sum_{x_2} \cdots \sum_{x_n} p(x_1, \dots, x_n) = 1$$

Reasoning

- Each outcome corresponds to a specific tuple.
- Summing over all tuples must give total probability = 1.

3. Joint PDF (Continuous Case)

Definition

If variables are continuous:

$$f_{X_1, \dots, X_n}(x_1, \dots, x_n) = \frac{\partial^n}{\partial x_1 \cdots \partial x_n} F_{X_1, \dots, X_n}(x_1, \dots, x_n)$$

Probability Computation

$$P(a_i \leq X_i \leq b_i, \forall i) = \int_{a_1}^{b_1} \cdots \int_{a_n}^{b_n} f(x_1, \dots, x_n) dx_1 \cdots dx_n$$

Properties

1. $f(x_1, \dots, x_n) \geq 0$
2. Total integral:

$$\int \cdots \int f(x_1, \dots, x_n) dx_1 \cdots dx_n = 1$$

4. Example-1 (Joint Distribution Construction)

Typical reasoning (as per slide structure)

1. Identify all random variables involved.

2. Define their joint outcomes.
 3. Construct:
 - Joint PMF (if discrete), or
 - Joint PDF (if continuous)
 4. Verify normalization condition.
-

5. Application: Large MIMO System

Interpretation from slide

- Multiple antennas $\rightarrow$ multiple random variables.
- Signal vectors are modeled as **N-dimensional random vectors**.

Reasoning

1. Each antenna output = one random variable.
 2. Entire system $\rightarrow$ joint distribution.
 3. Joint PDF/PMF captures:
 - Correlation
 - Dependence structure
-

6. Independence of Random Variables

Definition

Random variables $X_1, \dots, X_n$ are **independent** if:

Discrete case

$$p(x_1, \dots, x_n) = \prod_{i=1}^n p_{X_i}(x_i)$$

Continuous case

$$f(x_1, \dots, x_n) = \prod_{i=1}^n f_{X_i}(x_i)$$

Step-by-step reasoning

1. Start from joint probability.
 2. If it factorizes into marginals $\rightarrow$ independence.
 3. Otherwise $\rightarrow$ dependence exists.
-

7. Example-2: Uniformly Distributed 3D Vector

Setup

Let (X, Y, Z) be uniformly distributed in a region.

Joint PDF

$$f(x, y, z) = \begin{cases} \frac{1}{V}, & (x, y, z) \in \text{region} \\ 0, & \text{otherwise} \end{cases}$$

where V = volume of region.

Reasoning

1. Uniform $\rightarrow$ equal probability density.
2. Total probability = 1:

$$\int_V \frac{1}{V} dV = 1$$

3. Therefore density must be $1/V$.

8. Example-3: Trivariate Random Variables

General Framework

For three variables (X, Y, Z) :

Joint CDF

$$F(x, y, z) = P(X \leq x, Y \leq y, Z \leq z)$$

Joint PDF

$$f(x, y, z) = \frac{\partial^3}{\partial x \partial y \partial z} F(x, y, z)$$

Marginal Distributions

$$f_X(x) = \iint f(x, y, z) dy dz$$

$$f_Y(y) = \iint f(x, y, z) dx dz$$

$$f_Z(z) = \iint f(x, y, z) dx dy$$

Reasoning

1. Joint captures full dependence.
2. Marginals obtained by integrating out other variables.
3. Used to analyze individual behavior.

9. Summary of Key Concepts

- Joint distribution generalizes probability to multiple variables.
- Three representations:
 - Joint CDF
 - Joint PMF (discrete)
 - Joint PDF (continuous)
- Independence is verified via factorization.
- Applications include multi-dimensional systems such as MIMO.
- Marginalization reduces dimension.

Exam-Focused Insights

- Always start from **joint definition**.
- Verify **normalization** explicitly.
- For independence:
 - Show factorization step clearly.
- For continuous cases:
 - Use **multiple integrals carefully**.
- For vector distributions:
 - Identify region and compute volume correctly.

If you want, I can convert this into **short revision notes (1-page cheat sheet)** or **practice problems with solutions** based on this lecture.

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Where slide content is not readable, only **standard definitions directly corresponding to slide headings** are used, without introducing extra topics.

All reasoning is written **step-by-step**, as required for **closed-notes exam revision**.

Lecture 19 Scribe: N-Variate Joint PDF / PMF / CDF

Course: CSE400 – Fundamentals of Probability in Computing

Lecture: 19

Topic: Joint Distribution of N Random Variables

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1. Joint Distribution of N Random Variables

Definition

Let $X_1, X_2, \dots, X_n$ be n random variables defined on the same probability space.

The **joint cumulative distribution function (CDF)** is defined as:

$$F_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n) = P(X_1 \leq x_1, X_2 \leq x_2, \dots, X_n \leq x_n)$$

Step-by-step reasoning

1. Consider each random variable individually.
2. Form the joint event:

$$(X_1 \leq x_1) \cap (X_2 \leq x_2) \cap \dots \cap (X_n \leq x_n)$$

3. The joint CDF is the probability of this combined event.
4. This extends the univariate CDF to multiple dimensions.

2. Example-1 (Joint Distribution)

Reasoning procedure (as implied by slides)

1. Identify the set of random variables involved.
2. Determine all possible joint outcomes.
3. Assign probabilities (discrete) or density (continuous).
4. Ensure total probability equals 1:
 - Sum = 1 (discrete)

- Integral = 1 (continuous)

3. Application: Large MIMO System

From the slide reference L19 :

Interpretation

- A **Large MIMO system** involves multiple antennas.
- Each antenna output corresponds to a random variable.

Step-by-step reasoning

1. Let each antenna signal be X_i .
2. The system output is a vector:

$$(X_1, X_2, \dots, X_n)$$

3. The full system behavior is described by the **joint distribution**.
4. This captures:
 - Correlation between signals
 - Dependence structure

4. Joint PMF, CDF, PDF

4.1 Joint PMF (Discrete Case)

$$p(x_1, \dots, x_n) = P(X_1 = x_1, \dots, X_n = x_n)$$

Reasoning

1. Each tuple represents one outcome.
2. Probability assigned to each tuple.
3. Total probability must satisfy:

$$\sum_{x_1} \cdots \sum_{x_n} p(x_1, \dots, x_n) = 1$$

4.2 Joint CDF

$$F(x_1, \dots, x_n) = P(X_1 \leq x_1, \dots, X_n \leq x_n)$$

Reasoning

1. Convert equality events into inequality events.

2. Aggregate probabilities up to given thresholds.
3. Ensures monotonicity in each variable.

4.3 Joint PDF (Continuous Case)

$$f(x_1, \dots, x_n) = \frac{\partial^n}{\partial x_1 \cdots \partial x_n} F(x_1, \dots, x_n)$$

Reasoning

1. Start from joint CDF.
2. Differentiate with respect to all variables.
3. This yields density at a point.

Probability computation

$$P(a_i \leq X_i \leq b_i) = \int_{a_1}^{b_1} \cdots \int_{a_n}^{b_n} f(x_1, \dots, x_n) dx_1 \cdots dx_n$$

5. Independence of Random Variables

Definition

Random variables are independent if:

Discrete case

$$p(x_1, \dots, x_n) = \prod_{i=1}^n p_{X_i}(x_i)$$

Continuous case

$$f(x_1, \dots, x_n) = \prod_{i=1}^n f_{X_i}(x_i)$$

Step-by-step reasoning

1. Start with the joint distribution.
2. Compare it with the product of marginals.
3. If equal $\rightarrow$ independence holds.
4. If not $\rightarrow$ variables are dependent.

6. Example-2: Uniformly Distributed 3D Vector

Setup

Let (X, Y, Z) be uniformly distributed over a region.

Joint PDF

$$f(x, y, z) = \begin{cases} \frac{1}{V}, & (x, y, z) \in \text{region} \\ 0, & \text{otherwise} \end{cases}$$

Step-by-step reasoning

1. Uniform distribution implies constant density.
2. Let density = c .
3. Total probability must be 1:

$$\int_{\text{region}} c \, dV = 1$$

4. Hence:

$$c = \frac{1}{V}$$

7. Example-3: Trivariate Random Variables

Joint CDF

$$F(x, y, z) = P(X \leq x, Y \leq y, Z \leq z)$$

Joint PDF

$$f(x, y, z) = \frac{\partial^3}{\partial x \partial y \partial z} F(x, y, z)$$

Marginal Distributions

$$f_X(x) = \iint f(x, y, z) \, dy \, dz$$

$$f_Y(y) = \iint f(x, y, z) \, dx \, dz$$

$$f_Z(z) = \iint f(x, y, z) \, dx \, dy$$

Step-by-step reasoning

1. Start from joint PDF.
2. Integrate out unwanted variables.

3. Result is marginal distribution of remaining variable.

8. Overall Logical Flow (From Slides)

1. Define joint distribution for n variables.
2. Extend to:
 - PMF (discrete)
 - PDF (continuous)
 - CDF (general)
3. Study independence via factorization.
4. Apply to:
 - Engineering systems (MIMO)
5. Solve structured examples:
 - Uniform 3D distribution
 - Trivariate case

Exam-Oriented Key Points

- Always begin with **joint definition**.
- Clearly distinguish:
 - PMF vs PDF vs CDF
- For independence:
 - Show **product factorization explicitly**
- For continuous problems:
 - Use **multi-dimensional integration carefully**
- For uniform distributions:
 - Derive constant using **normalization condition**

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Where slide text is not readable, **only minimal standard definitions directly implied by headings** are used.

All reasoning is written **step-by-step in an exam-oriented format**, as requested.

Lecture 19 Scribe: N-Variate Joint PDF / PMF / CDF

Course: CSE400 – Fundamentals of Probability in Computing

Lecture: 19

Topic: Joint Distribution of N Random Variables

^{L19}

1. Joint Distribution of N Random Variables

Definition

Let $X_1, X_2, \dots, X_n$ be n random variables.

The **joint cumulative distribution function (CDF)** is:

$$F(x_1, x_2, \dots, x_n) = P(X_1 \leq x_1, X_2 \leq x_2, \dots, X_n \leq x_n)$$

Step-by-step reasoning

1. Start with multiple random variables defined on the same probability space.
2. Construct the joint event:

$$(X_1 \leq x_1) \cap (X_2 \leq x_2) \cap \dots \cap (X_n \leq x_n)$$

3. Evaluate the probability of this intersection.
4. This defines the joint CDF.

2. Example-1

Procedure (as implied by slide structure)

1. Identify the random variables.
2. Define the joint outcome space.
3. Assign probabilities (or density).

4. Verify normalization:

- Discrete: sum = 1
- Continuous: integral = 1

3. Application: Large MIMO System

Interpretation

- Multiple antennas correspond to multiple random variables.
- The system is represented as an n -dimensional random vector.

Step-by-step reasoning

1. Let each antenna output be X_i .
2. Form the vector:

$$(X_1, X_2, \dots, X_n)$$

3. The system behavior depends on the **joint distribution**.
4. This captures relationships among variables.

4. Joint PMF, CDF, PDF

4.1 Joint PMF

$$p(x_1, \dots, x_n) = P(X_1 = x_1, \dots, X_n = x_n)$$

Step-by-step reasoning

1. Each tuple $(x_1, \dots, x_n)$ represents one outcome.
2. Assign probability to each tuple.
3. Ensure:

$$\sum_{x_1} \cdots \sum_{x_n} p(x_1, \dots, x_n) = 1$$

4.2 Joint CDF

$$F(x_1, \dots, x_n) = P(X_1 \leq x_1, \dots, X_n \leq x_n)$$

Step-by-step reasoning

1. Convert values into cumulative inequalities.

2. Compute probability of joint event.
3. Function increases as any x_i increases.

4.3 Joint PDF

$$f(x_1, \dots, x_n) = \frac{\partial^n}{\partial x_1 \cdots \partial x_n} F(x_1, \dots, x_n)$$

Step-by-step reasoning

1. Begin with joint CDF.
2. Differentiate with respect to each variable.
3. Obtain density function.

Probability evaluation

$$P(a_i \leq X_i \leq b_i) = \int_{a_1}^{b_1} \cdots \int_{a_n}^{b_n} f(x_1, \dots, x_n) dx_1 \cdots dx_n$$

5. Independence of Random Variables

Definition

Random variables are independent if:

$$p(x_1, \dots, x_n) = \prod_{i=1}^n p_{X_i}(x_i)$$

or

$$f(x_1, \dots, x_n) = \prod_{i=1}^n f_{X_i}(x_i)$$

Step-by-step reasoning

1. Compute the joint distribution.
2. Compute marginal distributions.
3. Multiply marginals.
4. Compare with joint:
 - Equal $\rightarrow$ independent
 - Not equal $\rightarrow$ dependent

6. Example-2: Uniformly Distributed 3D Vector

Setup

Let (X, Y, Z) be uniformly distributed in a region.

Joint PDF

$$f(x, y, z) = \begin{cases} c, & \text{inside region} \\ 0, & \text{outside} \end{cases}$$

Step-by-step reasoning

1. Uniform distribution $\Rightarrow$ constant density c .
2. Use normalization:

$$\int_{\text{region}} c \, dV = 1$$

3. Solve:

$$c = \frac{1}{\text{Volume of region}}$$

7. Example-3: Trivariate Random Variables

Joint CDF

$$F(x, y, z) = P(X \leq x, Y \leq y, Z \leq z)$$

Joint PDF

$$f(x, y, z) = \frac{\partial^3}{\partial x \partial y \partial z} F(x, y, z)$$

Marginal Distributions

$$f_X(x) = \iint f(x, y, z) \, dy \, dz$$

$$f_Y(y) = \iint f(x, y, z) \, dx \, dz$$

$$f_Z(z) = \iint f(x, y, z) \, dx \, dy$$

Step-by-step reasoning

1. Start from joint PDF.
2. Integrate out remaining variables.

3. Obtain marginal of interest.

8. Logical Flow of Lecture (As per Slides)

1. Define joint distribution for n variables.

2. Extend to:

- PMF (discrete)
- PDF (continuous)
- CDF (general)

3. Study independence.

4. Apply to:

- MIMO systems

5. Solve examples:

- Uniform 3D vector
- Trivariate case

Exam-Focused Summary

- Joint distribution describes multiple variables together.
- Use:
 - PMF $\rightarrow$ discrete
 - PDF $\rightarrow$ continuous
 - CDF $\rightarrow$ general
- Independence requires **factorization into marginals**.
- Marginals obtained via **integration/summation**.
- Uniform distributions require **normalization to find constant density**.

This scribe is **strictly aligned with the PPT outline and structure** and formatted for **closed-notes exam revision with explicit reasoning steps**. L19

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Lecture 19 Scribe: N-Variate Joint PDF / PMF / CDF

Course: CSE400 – Fundamentals of Probability in Computing

Lecture: 19

Topic: Joint Distribution of N Random Variables

L19

1. Joint Distribution of N Random Variables

Definition

Let $X_1, X_2, \dots, X_n$ be n random variables defined on a common probability space.

The **joint cumulative distribution function (CDF)** is defined as:

$$F_{X_1, \dots, X_n}(x_1, \dots, x_n) = P(X_1 \leq x_1, X_2 \leq x_2, \dots, X_n \leq x_n)$$

Step-by-step reasoning

1. Consider each random variable individually.
2. Form the joint event:

$$(X_1 \leq x_1) \cap (X_2 \leq x_2) \cap \dots \cap (X_n \leq x_n)$$

3. Compute the probability of this intersection.
4. This defines the joint CDF for n variables.

2. Example-1

Reasoning procedure

1. Identify all involved random variables.
2. Construct the joint outcome space.
3. Assign probability values (PMF) or density (PDF).

4. Verify normalization:

- Discrete case:

$$\sum p(x_1, \dots, x_n) = 1$$

- Continuous case:

$$\int f(x_1, \dots, x_n) dx_1 \cdots dx_n = 1$$

3. Application: Large MIMO System

Interpretation (from slide reference)

- Multiple antennas correspond to multiple random variables.
- The system is modeled as an n -dimensional random vector.

Step-by-step reasoning

1. Let each antenna output be X_i .
2. Form the vector:

$$(X_1, X_2, \dots, X_n)$$

3. The behavior of the system is described using the joint distribution.
4. The joint model captures relationships among variables.

4. Joint PMF, CDF, PDF

4.1 Joint PMF (Discrete Case)

$$p(x_1, \dots, x_n) = P(X_1 = x_1, \dots, X_n = x_n)$$

Step-by-step reasoning

1. Each tuple corresponds to a specific outcome.
2. Assign a probability to each tuple.
3. Ensure total probability:

$$\sum_{x_1} \cdots \sum_{x_n} p(x_1, \dots, x_n) = 1$$

4.2 Joint CDF

$$F(x_1, \dots, x_n) = P(X_1 \leq x_1, \dots, X_n \leq x_n)$$

Step-by-step reasoning

1. Convert point probabilities into cumulative form.
2. Evaluate probability of joint inequalities.
3. Observe that the function is non-decreasing in each argument.

4.3 Joint PDF (Continuous Case)

$$f(x_1, \dots, x_n) = \frac{\partial^n}{\partial x_1 \cdots \partial x_n} F(x_1, \dots, x_n)$$

Step-by-step reasoning

1. Begin with joint CDF.
2. Differentiate with respect to all variables.
3. Obtain density function.

Probability computation

$$P(a_i \leq X_i \leq b_i) = \int_{a_1}^{b_1} \cdots \int_{a_n}^{b_n} f(x_1, \dots, x_n) dx_1 \cdots dx_n$$

5. Independence of Random Variables

Definition

Random variables $X_1, \dots, X_n$ are independent if:

$$p(x_1, \dots, x_n) = \prod_{i=1}^n p_{X_i}(x_i)$$

or

$$f(x_1, \dots, x_n) = \prod_{i=1}^n f_{X_i}(x_i)$$

Step-by-step reasoning

1. Compute the joint distribution.
2. Compute marginal distributions.
3. Multiply marginals.
4. Compare:
 - If equal $\rightarrow$ independence holds
 - Otherwise $\rightarrow$ dependence exists

6. Example-2: Uniformly Distributed 3D Vector

Setup

Let (X, Y, Z) be uniformly distributed in a region.

Joint PDF

$$f(x, y, z) = \begin{cases} c, & \text{inside region} \\ 0, & \text{outside} \end{cases}$$

Step-by-step derivation

1. Uniform distribution $\Rightarrow$ constant density c .
2. Apply normalization:

$$\int_{\text{region}} c \, dV = 1$$

3. Solve:

$$c = \frac{1}{\text{Volume of region}}$$

7. Example-3: Trivariate Random Variables

Joint CDF

$$F(x, y, z) = P(X \leq x, Y \leq y, Z \leq z)$$

Joint PDF

$$f(x, y, z) = \frac{\partial^3}{\partial x \partial y \partial z} F(x, y, z)$$

Marginal Distributions

$$f_X(x) = \iint f(x, y, z) \, dy \, dz$$

$$f_Y(y) = \iint f(x, y, z) \, dx \, dz$$

$$f_Z(z) = \iint f(x, y, z) \, dx \, dy$$

Step-by-step reasoning

1. Start with joint PDF.
 2. Integrate out unwanted variables.
 3. Obtain marginal distributions.
-

8. Logical Structure of the Lecture

From the outline ^{L19}:

1. Define joint distribution.
 2. Introduce PMF, CDF, PDF.
 3. Study independence.
 4. Apply to engineering system (MIMO).
 5. Solve examples:
 - General joint construction
 - Uniform 3D vector
 - Trivariate variables
-

Exam-Focused Key Points

- Joint CDF is the foundational definition.
 - PMF (discrete) and PDF (continuous) derive from it.
 - Independence requires **factorization into marginals**.
 - Marginals obtained via **integration or summation**.
 - Uniform distributions require **normalization to determine constant density**.
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This scribe is **fully aligned with the PPT content and structure** ^{L19} and formatted for **precise, closed-notes exam revision with explicit step-by-step reasoning**.

Sources